

Obligatory Slots

Under Reference-Free Evaluation, Dropping a Distinction the Source Never Made Is Free

They came.



வந்தார்கள் or வந்தன

Tamil must choose. English never said.

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The whole paper in four sentences

1

Tamil forces a choice English never makes.

The verb must say whether the subject is a person, and for one person usually whether male or female. English "they came" says neither.

2

We built a test set to study that choice.

TamilLingBench: English→Tamil test sentences plus a rule-based checker that reads the Tamil verb ending automatically.

3

Metrics that use a reference translation do punish a wrong ending.

We had predicted that AI-based (neural) metrics would not. That prediction was wrong, and we report it openly.

4

Metrics that use no reference do not.

On real system errors they prefer the wrong form more often than the right one, and skipping the choice costs nothing.

How this presentation is organised

1

The problem

What an "obligatory slot" is, with examples

2

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4

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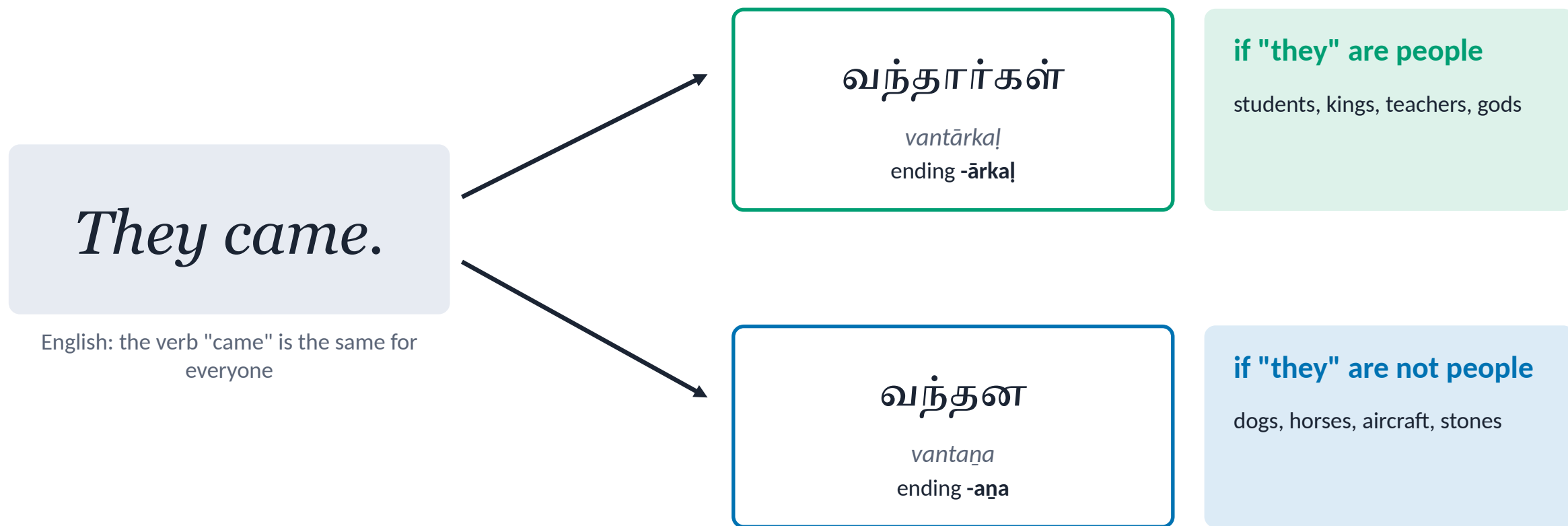


1

The problem

When the target language must say something the source language never said.

One English sentence, two possible Tamil translations



The English never says which. **Tamil cannot avoid saying it.** A translation system has to pick one.

Why? The Tamil verb agrees with its subject

Tamil verb	How it sounds (romanised)	English	What the ending says about the subject
வந்தான்	<i>vantāṇ</i>	he came	one person, masculine
வந்தாள்	<i>vantāḷ</i>	she came	one person, feminine
வந்தார்	<i>vantār</i>	he / she came	one person, respectful; does not say male or female
வந்தார்கள்	<i>vantārkaḷ</i>	they came	several people (also: one person, very respectfully)
வந்தது	<i>vantatu</i>	it came	one animal or thing
வந்தன	<i>vantaṇa</i>	they came	several animals or things

Six Tamil words, **one** English word "came". Every Tamil sentence about a third person (he, she, it, they) has to pick one of them.

Three things the Tamil verb must mark

Rationality (திணை tiṇai)

Is the subject a person (or a god), or not?

மாணவர்கள் வந்தார்கள்

The students came.

நாய்கள் வந்தன

The dogs came.

English: **not marked at all**. Only the noun hints at it.

Gender

Is one person male or female?

அறம் வந்தான்

Aram (a boy) came.

அறம் வந்தாள்

Aram (a girl) came.

English: only through names or he/she.

Number (our control)

One or many?

மாடு நகர்ந்தது

The cow moved.

மாடுகள் நகர்ந்தன

The cows moved.

English: **clearly marked** (cow / cows).

Rationality (திணை tiṇai), in simple words

உயர்திணை **uyartiṇai** ("high class")

People and gods. The plural verb ends in **-ārkaḷ**.

The students came at dawn.

மாணவர்கள் வந்தார்கள்.

māṇavarkaḷ vantārkaḷ.

அஃறிணை **akriṇai** ("not-high class")

Animals, plants and objects. The plural verb ends in **-aṇa**.

The dogs came at dawn.

நாய்கள் வந்தன.

nāykaḷ vantaṇa.

English grammar has **no marker** for this. The noun ("students" or "dogs") only lets you **guess** it from meaning.

The key idea: an "obligatory slot"

Definition. A slot is **obligatory** when every correct Tamil translation must fill it, even though the English does not say what to fill it with. What fills it is called its **value** (for example "people" or "not people").



Ordinary translation error

The information **is** in the source and the system loses it. Example: English "she" translated as masculine.

Obligatory-slot situation

The information is **not** in the source at all, but the target language still demands it.

There is **nothing to copy** from the English, yet a value **must appear** in the Tamil.

Why it matters: this is not a matter of style

Dehumanising

- Putting a **person** in the non-people class treats them like an animal or an object.
- Every Tamil verb about a third person forces this choice.

Misgendering

- Choosing masculine for a woman, or feminine for a man, is a real harm.
- This is why gender bias in translation is widely studied.

Invisible to the reader

- The Tamil reader usually **cannot see** the English source.
- They have **no way to tell** the error came from the system, not the speaker.

Example: "The students came."

Same sentence, same verb. Only the ending differs, and one ending treats the students as animals or objects.

மாணவர்கள் வந்தார்கள்.

māṇavarkaḷ vantārkaḷ.

✓ correct: people ending

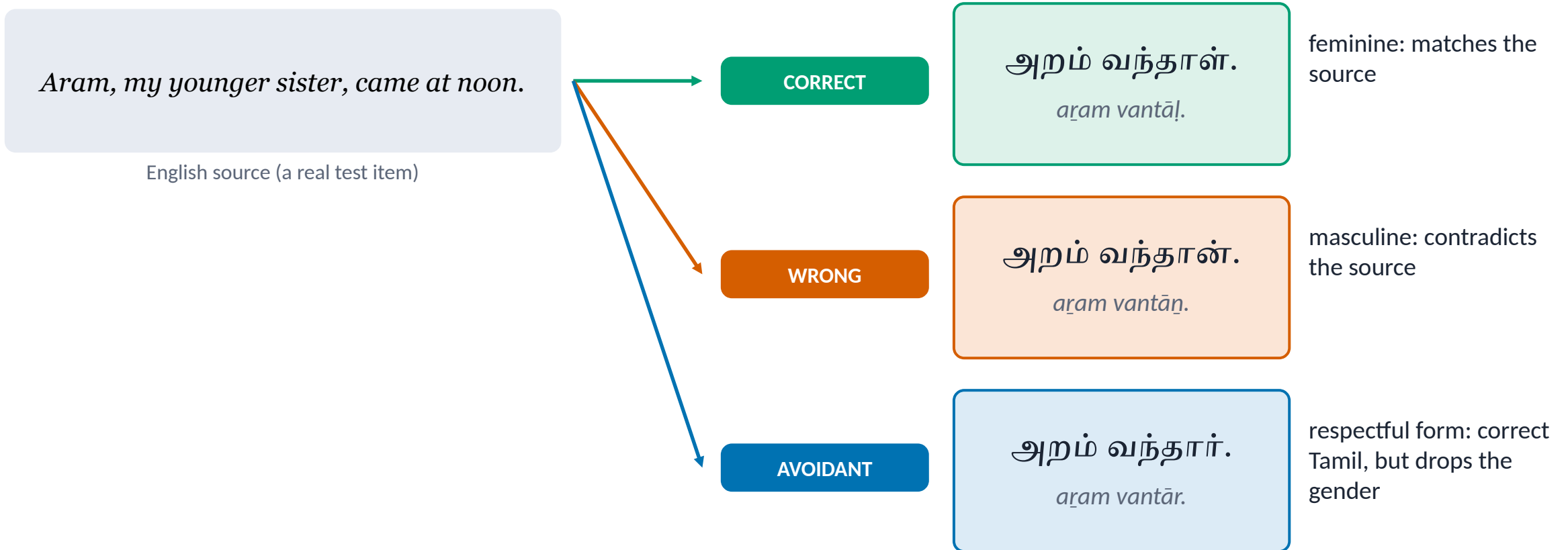
மாணவர்கள் வந்தன.

māṇavarkaḷ vantaṇa.

✗ dehumanising: non-people ending

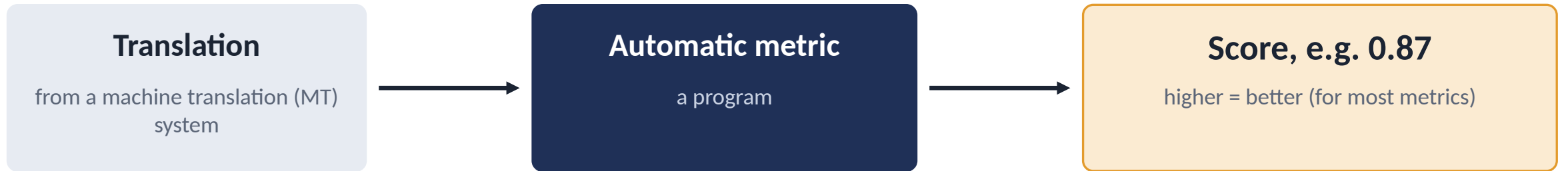
A translation can be fluent, accurate in every other word, highly scored, and still **socially unusable**.

What can a translation system do? Three possible outcomes



We count **avoiding** as its own outcome, **never as wrong**. That choice makes the paper's main result visible.

How do people judge translation quality? Automatic metrics



Research

Compare systems in papers and in open competitions ("shared tasks") such as WMT.

Choosing a system

Decide which system to deploy or which output to keep.

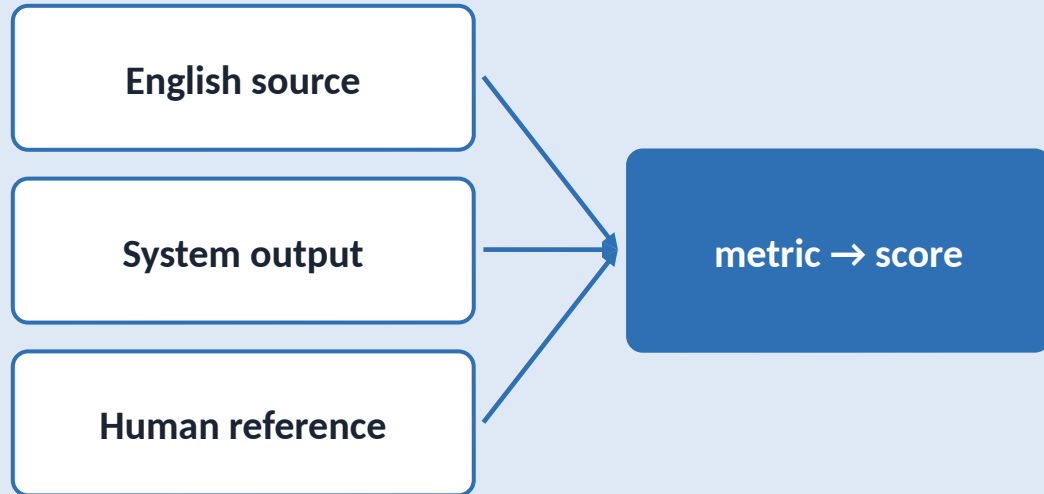
Monitoring

Watch the quality of a live system, where no human checks each sentence.

If a metric **cannot see** an error, that error is invisible everywhere the metric is used.

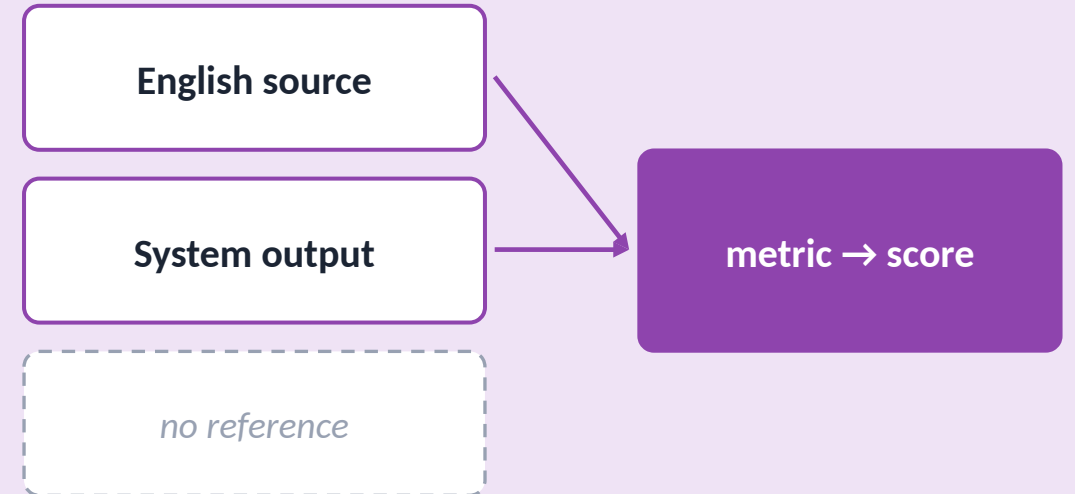
Two kinds of metrics: with a reference, and without

Reference-based



COMET-22 · xCOMET-XL · MetricX-24 · MetricX-23 · chrF2 · chrF++

Reference-free (quality estimation, "QE")



CometKiwi-22 · MetricX-24-QE

In real use there is **usually no human reference**, e.g. monitoring a live system. That is where reference-free metrics are used.

Our research questions

When the Tamil verb must choose, and the system chooses **wrong**, or **does not choose at all**, does automatic evaluation notice?

Q1 • Metrics (main question)

How much does each metric lower its score for a wrong or missing value?
Sections 5–6.

Q2 • Translation systems

How often do 13 real systems choose, choose correctly, or avoid? Section 7.

Q3 • Inside a model

Inside the neural network that translates: at what layer is the choice settled? Section 8.

2

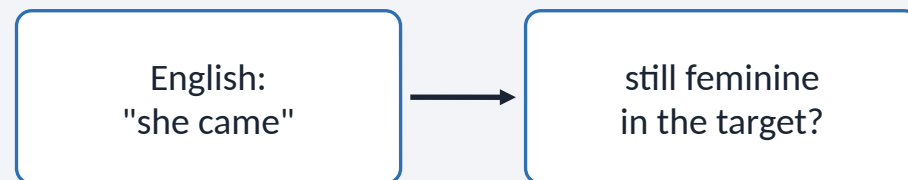
What earlier work did, and what was missing

Many test sets exist. None of them asks our question.

Earlier test sets check whether information is carried across

Test set	Year	What it checks
LingEval97	2017	agreement, negation and more (English→German)
Bawden et al.	2018	links between sentences, using context
ContraPro	2018	pronouns that need context
Morpheval	2017–18	word endings: does a feature shown in the source survive?
DFKI suites	WMT yearly	many kinds of grammar problems

The shared idea: **propagation**



The feature is **marked** in the source: shown by a word or a word ending. The test asks whether the system **keeps** it.

All of these test information that the source **does contain**. Our slots have **nothing to carry across**.

Our case is different: there is nothing to copy

Morpheval-style test (propagation)

- Source **marks** the feature: "**she** came".
- Question: is it **kept** in the translation?
- There is always a right answer, taken from the source.

Obligatory slot (this paper)

- Source **does not mark** it: "**they** came".
- Question: what does the system put in the slot, and **does evaluation notice** when it is wrong or missing?
- Sometimes there is **no right answer** in the source at all.

Propagation asks: **was the information kept?** We ask: **what happens when there was no information to keep?**

The closest earlier work, and how we differ

Work	What it does	What it does not do
GENDER1PERSON (WMT 2025)	Measures which gender systems choose when English "I" is unmarked (Russian, Serbian)	Only gender; does not ask what metrics charge
CoCoA-MT, IWSLT formality (2022–23)	Formal vs informal "you"; the unmarked case appears	Only as a side baseline, not studied as a condition
Měchura (2022)	Names the category of "bias-causing ambiguities"	Measures no system
GRAPE, Prates et al., others	How often systems pick each gender, measured in several ways	Gender only; not about metrics

No earlier study asked **how much metrics lower the score** for these errors, and we found **no earlier machine-translation study of rationality**.

Test sets for metrics exist, but not for this

DEMETER (2022)

Plants errors in translations to see whether metrics notice. **All its examples are into English.**

ACES (2025)

The largest test set for metrics. It leaves out word-ending errors on purpose. Only **1 of its 36,476** examples is English→Tamil.

IndicMT Eval (2023)

Covers Tamil. Metrics struggle with **fluency** errors (sentences that read unnaturally), which is where our errors land.

No metric test set covers English→Tamil verb agreement, and none studies an **unmarked** distinction (one the English does not show).

Standard Tamil language data cannot even label these features

0

words labelled feminine in UD Tamil-TTB, a standard grammar-labelled Tamil collection (a "treebank")

0

verbs with the he / she endings -ஆன் / -ஆள் in that collection

0

words labelled for inclusive vs exclusive "we" in either Tamil treebank

41 / 41

negative verb forms wrongly labelled as positive

7

central grammar categories the standard labels cannot express

absent

Tamil in UniMorph, the big multilingual word-form database

So we had to **build our own test set and checker.**

The gap, in one table

	Feature absent from source	Tamil / Dravidian	Rationality studied	Asks what metrics charge
Propagation test sets (ContraPro, Morpheval, ...)	✗	✗	✗	✗
Gender-skew studies (GENDER1PERSON, GRAPE, ...)	✓	✗	✗	✗
Formality tasks (CoCoA-MT, IWSLT)	partly	✗	✗	✗
Metric test sets (DEMETR, ACES)	✗	partly	✗	✓
This paper	✓	✓	✓	✓

We found **no earlier study** of **what evaluation charges** when a translation must fill a slot the source left **empty**.

What this paper contributes

1

A measurement of what metrics charge

for a wrong or missing value, including our failed prediction

2

TamilLingBench

a generator of test sentences + a rule-based Tamil checker, 12 automatic checks, reviewed by 2 native Tamil speakers

3

Three flaws in our own first release

and the automated audit that caught them

4

A pre-registered test inside one model

where the decision is made, reported with the part that failed

5

Measurements of 13 translation systems

preliminary, plus facts about gaps in Tamil language data

3

Our test set: TamilLingBench

Which slots, which conditions, how items are made, and how Tamil is checked automatically.

What we test: three slots, three different roles

Slot	Tamil forms tested	Earlier MT research	Role in this paper
Rationality (திணை)	வந்தார்கள் / வந்தன (vantārkaḷ / vantaṇa)	none found	new; results provisional
Gender	வந்தான் / வந்தாள் / வந்தார் (vantāṇ / vantāḷ / vantār)	heavily studied	comparison with earlier gender research
Number	வந்தது / வந்தன (vantatu / vantaṇa)	marked in both languages	control: should be easy

Removed: polite vs informal "you" (நீ / நீங்கள்) and inclusive vs exclusive "we" (நாம் "we, including you" / நாங்கள் "we, not you")

Tested in earlier versions, then removed because their test sentences were badly built. No results are reported for them.

What a "control" is, and why number is ours

Number: English **does** mark it

The cow moved outside the gate.

மாடு நகர்ந்தது.
māṭu nakarntatu.

The cows moved outside the gate.

மாடுகள் நகர்ந்தன.
māṭukaḷ nakarntaṇa.

- A **control** is a part of the experiment where you already know the expected answer.
- If systems fail the control, suspect the **test**, not the systems.
- Number is marked in English, so a good system should be near perfect on it.
- **Honest note:** on these items the number control did not work cleanly, so the paper draws **no comparison between slots**.

Four conditions: where is the clue?

Instructed

The instruction names the Tamil form to use.

Can the system produce it at all?
(its best case)

gold ✓

In-sentence cue

The English sentence itself contains the clue ("the dogs").

A normal test with a right answer

gold ✓

Prior-sentence cue

The clue is only in the sentence before.

Uses context; answers balanced
so guessing = 50%

gold ✓

Cue-free

No clue anywhere. **No correct answer exists.**

We record what systems choose
by default

no gold

A **cue** is simply a clue. **Gold** is the correct answer. Only **cue-free** items have no gold: that is where the forced choice happens.

Example: one rationality item in all four conditions

Condition	English given to the system ([...] = context, not translated)	Tamil gold (correct answer)
Instructed	<i>They came at dawn. + instruction: "...realising the rationality value 'UYAR' on the verb."</i>	வந்தார்கள். <i>vantārkaḷ.</i>
In-sentence	<i>The dogs came at dawn.</i>	நாய்கள் வந்தன. <i>nāykaḷ vantaṇa.</i>
Prior-sentence	<i>[The horses had gathered outside since dawn.] They came at dawn.</i>	வந்தன. <i>vantaṇa.</i>
Cue-free	<i>[The sky had been overcast since dawn.] They came at dawn.</i>	<i>no gold: any value is acceptable</i>

Same sentence pattern, same verb. Only the **location of the clue** changes. Real items from the released TamillIngBench file.

Examples: gender and number

Gender

Cue	English	Tamil gold
in-sentence	Aram, my younger sister, came at noon.	அறம் வந்தாள். aṛam vantaḷ.
in-sentence	Aram, my younger brother, came at noon.	அறம் வந்தான். aṛam vantaṅ.
cue-free	[Aram, our new neighbour, had been waiting at the station since noon.] Aram came at noon.	<i>no gold; systems mostly say வந்தார் (avoid)</i>

Number (control)

Cue	English	Tamil gold
in-sentence	The cows moved outside the gate.	மாடுகள் நகர்ந்தன. māṭukaḷ nakarntaṇa.
in-sentence	The cow moved outside the gate.	மாடு நகர்ந்தது. māṭu nakarntatu.
cue-free	[A new tower had been opened the previous week.] The aircraft moved outside the gate.	<i>no gold: "aircraft" can be one or many</i>

Important: only the cue-free condition has no right answer

3 conditions with gold: Instructed · In-sentence · Prior-sentence

These work like normal test sets: there is a correct answer, taken from the instruction, the sentence, or the sentence before.

1 condition without gold: Cue-free

The genuinely new part. We report what systems choose.

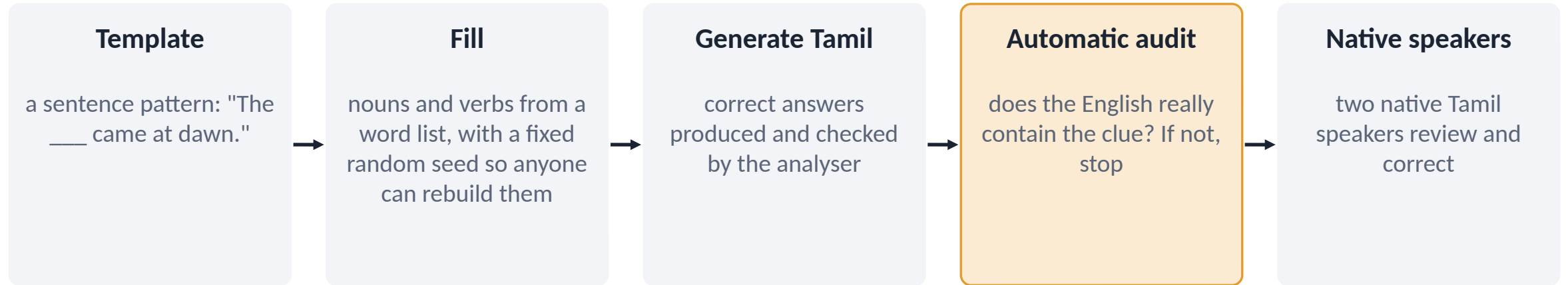
✓ The main comparison

In-sentence **vs** cue-free: same sentence pattern, same kind of clue; the clue is simply **deleted**.

⚠ In-sentence vs prior-sentence: rationality only

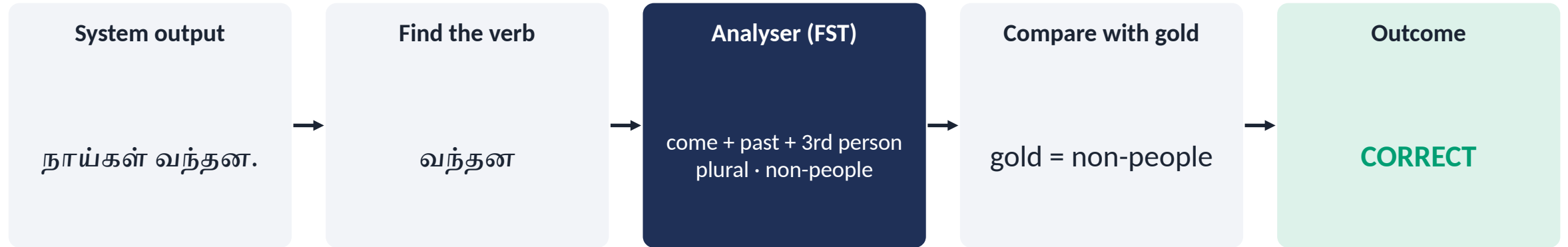
Allowed only where both give the clue **the same way** (rationality: a noun in both). In **gender** one uses a Tamil name, the other an English word, so **no gender comparison** is made.

How the test items were made



- **2,873 test items** in the release (an item = one test sentence plus its correct answer).
- Some items are **hidden** (a "blind" set). We hide **whole templates**, not single sentences, because anyone with the public generator could rebuild a hidden sentence.
- **12 automatic checks** run every time the test set is rebuilt.

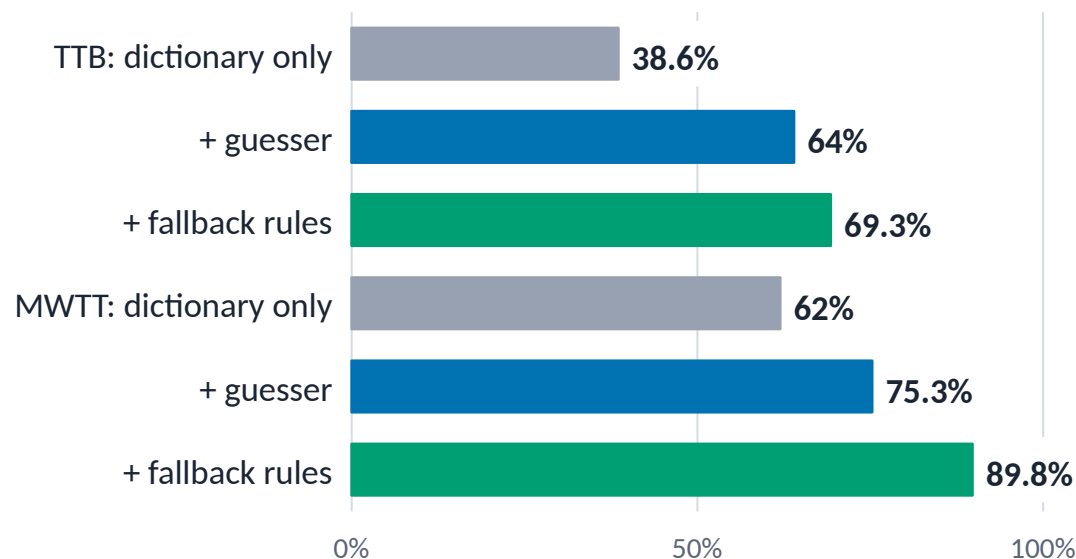
Checking Tamil automatically: the morphological analyser



- **FST = finite-state transducer**: a rule-based word analyser built from a dictionary and grammar rules.
- **Why not an AI (neural) model?** Rules give the **same answer every time**, and you can see why. The best published AI tool for Tamil grammar errors scores only **73.4 out of 100** (F1 score).
- Built on **ThamizhiMorph** (Sarveswaran et al., 2021) plus hand-written fallback rules.

The analyser is good, but not perfect, and we say so

Share of verbs from two standard Tamil treebanks the analyser can read



Hand-checked on masculine verbs

- Every reading it gave was right (**precision 100%**).
- It found **98.7%** of them (**recall**).

Cannot read it?

Labelled **UNPARSED**, reported separately, **never dropped**: 6.5% of our outputs on average. Our test sentences use a small set of pre-checked verbs; treebanks contain every kind of real verb.

The analyser is **not 100% accurate** and the paper **never claims** it is.

Five repairs we had to make to the analyser

1 -ஆர் -ār can mean one respected person **or** several people; resolved using context

2 -நீர்கள் / -உங்கள் second-person endings ambiguous in the same way

3 -அனர் -aṇar tagged singular although it is plural

4 sound change at word ends hid a pronoun and created a false verb reading

5 ஓடின **ōṭiṇa** "(they) ran" read only as a describing form ("that ran"), not a full verb, so **correct Tamil was marked wrong**. Fixed; everything re-scored.

Repairs change **how we read** the analyser's output. The published analyser itself is **untouched**.

Every output gets one of five outcomes

CORRECT

the value matches the gold answer

WRONG

the value is the other one

AVOIDANT

no value chosen: verb dropped, a neutral form (e.g. வந்தார்), or a negative sentence that removes the ending

UNDECIDABLE

the form is ambiguous even after using context

UNPARSED

the analyser could not read the output

We **never** count **AVOIDANT** as **wrong**. Otherwise a system that politely declines to guess would look as if it makes many mistakes.

Two numbers we report: commitment and accuracy



Commitment = 8 / 10 = 80%

Share of items where the system **chose a value** (correct + wrong). Avoidant and unparsed do not count as choices.

Accuracy = 6 / 8 = 75%

Among the items where it **did** choose, the share it got **right**. Computed only over committed items.

Keeping them apart shows when context makes systems **commit more** but **get it right less** (section 7). A **cell** = one system × one slot × one condition; if under 30 items in a cell had a choice, we print **n/a** (too few to trust).

4

The experiment

13 translation systems, fixed prompts, and a corpus baseline for the cue-free case.

First: what kind of translation systems are these?

Language model (LLM)

A large **neural network** trained on huge amounts of text. It continues text, and translates when asked.

Prompt

The instruction we give it, e.g. "**Translate the following English sentence into Tamil.**"

Parameters (1B, 4B, ...)

The numbers inside the network, learned in training. **4B $\approx$ 4 billion**: a rough measure of size.

Fine-tuning

Extra training on **one narrow task**, e.g. translation, starting from an existing model.

Dedicated MT system

A network built **only to translate**: reads English, writes Tamil. It needs **no prompt**.

Neural

Built from a **trained neural network**, rather than from hand-written rules.

The 13 translation systems we tested

Gemma 3 (Google)

1B

4B

12B

27B

scale series inside one family

Gemma 3n (Google)

E2B

E4B

efficient variants

Qwen3 (Alibaba)

8B

14B

32B

a second family, several sizes

Llama 3.1 (Meta)

8B

a third family

Sarvam-Translate

Gemma 3 4B, fine-tuned

translation fine-tuning vs its base model

NLLB-200 (Meta)

3.3B

dedicated MT, no prompt

IndicTrans2 (AI4Bharat)

1B

dedicated MT, no prompt



11 prompted language models



translation-tuned LLM (counted with the 11)



2 dedicated MT systems

Why these systems: three comparisons built in

Size

Gemma 3 at **1B, 4B, 12B, 27B**: does a bigger model behave differently?

Family

Qwen3 (three sizes) and **Llama 3.1**: do different model families behave differently at similar size?

Architecture

Dedicated MT (**NLLB-200, IndicTrans2**) vs prompted language models.

Bonus comparison: Sarvam-Translate vs Gemma 3 4B

Sarvam-Translate is Gemma 3 4B with extra translation training for Indian languages. Comparing the two shows what that **extra translation training alone** changes.

The goal is **comparisons**, not a leaderboard. No commercial frontier system was included, by decision.

How the systems were run

- **Greedy decoding** for the language models: always take the single most likely next word, so results are repeatable.
- **Beam search** for the two dedicated MT systems (it keeps several candidate translations at once and picks the best), at their standard settings.
- **37 runs**, fixed seed 20260807; each run records the exact model version, configuration and decoding settings.
- **We score the text each system actually writes.** Asking which of two answers it prefers would need hidden internal numbers that many systems do not share.
- An automatic check makes the build **fail** if any of the 13 systems is missing from any headline table.

Prompt wording changes the numbers a lot

Spread between prompts, over 73 cells

7.5

percentage points: the typical
(median) spread

82.5

percentage points: the worst
case

Prior-sentence prompt (used for every language model)

"Translate the sentence marked TRANSLATE into Tamil. The preceding sentence is given only as context; do not translate it. Output only the Tamil translation of the marked sentence."

So each headline number uses **one fixed prompt for all systems**. That makes systems comparable, but each number describes a system **with this prompt**, not the system in general.

The cue-free case: comparing with ordinary Tamil text

No gold, so no accuracy

Instead: **which value does the system pick**, and how often?

Compare with a "corpus prior"

How often each value appears in **three large collections of ordinary Tamil text** (corpora).

Always a range

The three collections disagree a lot, so we give a **range**, not one number: e.g. 34.5 to 69.6%.

What we can and cannot say

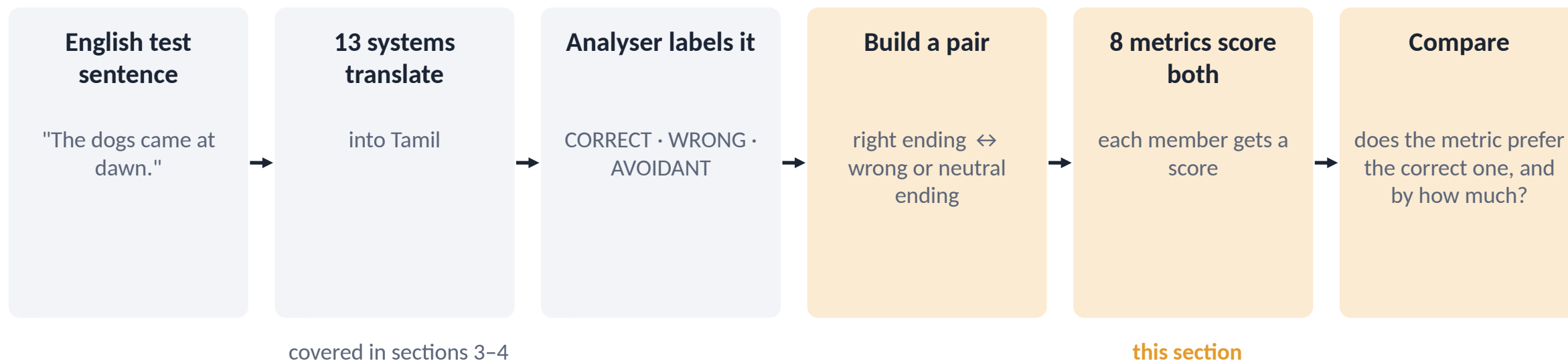
- ✓ "deviates from general Tamil text"
- ✗ "copies its own training data" (no system's training data was measured)

5

The metrics, and how we test them

Eight metrics, three kinds of evidence, three measurements, and a prediction written down in advance.

How everything fits together, from test sentence to metric score



The metrics are never shown the analyser's labels. They only see the sentences, and we check whether their **scores agree** with the labels.

The eight metrics we tested

Metric	Uses a reference?	Neural?	Score range	What it does, in one line
COMET-22	with reference	yes	0 – 1	neural model trained on scores humans gave to translations
xCOMET-XL	with reference	yes	0 – 1	like COMET, and also marks error spans with a severity
MetricX-24	with reference	yes	0 – 25	Google's neural metric (raw: lower is better)
MetricX-23	with reference	yes	0 – 25	the previous MetricX version
chrF2	with reference	no	0 – 100	counts shared character sequences with the reference
chrF++	with reference	no	0 – 100	chrF plus shared words and word pairs
CometKiwi-22	no reference (QE)	yes	0 – 1	COMET-style model that sees only source + output
MetricX-24-QE	no reference (QE)	yes	0 – 25	MetricX variant that sees only source + output

6 reference-based, 2 reference-free. 6 neural, 2 not neural (chrF2, chrF++). Score ranges differ, which matters when comparing numbers.

How each kind of metric can see the error

English: "The dogs came at dawn." System output: நாய்கள் வந்தார்கள். (people ending: wrong)

Reference-based metric

நாய்கள் வந்தன.

reference

நாய்கள் வந்தார்கள்.

output: differs from the reference in the verb → **visible**

Reference-free metric

The dogs came at dawn.

source

நாய்கள் வந்தார்கள்.

must **know by itself** that dogs take the non-people ending

A reference-free metric needs **real knowledge of Tamil grammar** to catch this error.

A real pair from our data, with real scores

English: "The dogs came at dawn."

அதிகாலையில் நாய்கள் வந்தன.

atikālaiyil nāykaḷ vantaṇa.

correct: non-people ending

அதிகாலையில் நாய்கள் வந்தார்கள்.

atikālaiyil nāykaḷ vantārkaḷ.

wrong: people ending

Metric	Score: correct	Score: wrong	Which does it prefer?
CometKiwi-22 (no reference)	0.882	0.885	the WRONG one
COMET-22 (with reference)	0.983	0.958	the correct one

One example to show the idea. In this constructed pair the reference **is** the correct sentence, which helps COMET-22. Real results use **thousands** of pairs.

Three kinds of evidence, which we call "arms"

Controlled

3,500 records

We **build** a minimal pair: same sentence, only the verb ending changed. The analyser confirms the changed one carries the wrong value.

நாய்கள் வந்தன.
↔ நாய்கள் வந்தார்கள்.

Natural

1,479 records

A mistake a system **really made**, paired with the same sentence with **only the ending corrected** (done automatically, checked by the analyser).

மருத்துவர்கள் ... வந்தது.
↔ மருத்துவர்கள் ... வந்தார்கள்.

Avoidance

1,252 records

The system used the **neutral form** instead of choosing a gender. Paired with the gendered version.

அறம் வந்தார்.
↔ அறம் வந்தாள்.

6,231 records in total. The arms can disagree, so every result names the arm it comes from.

A real mistake from the natural arm

English: "The doctors came by the old wall." System: NLLB-200 3.3B

மருத்துவர்கள் பழைய சுவர் மூலம் வந்தது.

maruttuvarkaḷ paḷaiya cuvar mūlam vantatu.

what the system wrote: non-people ending (wrong)

மருத்துவர்கள் பழைய சுவர் மூலம் வந்தார்கள்.

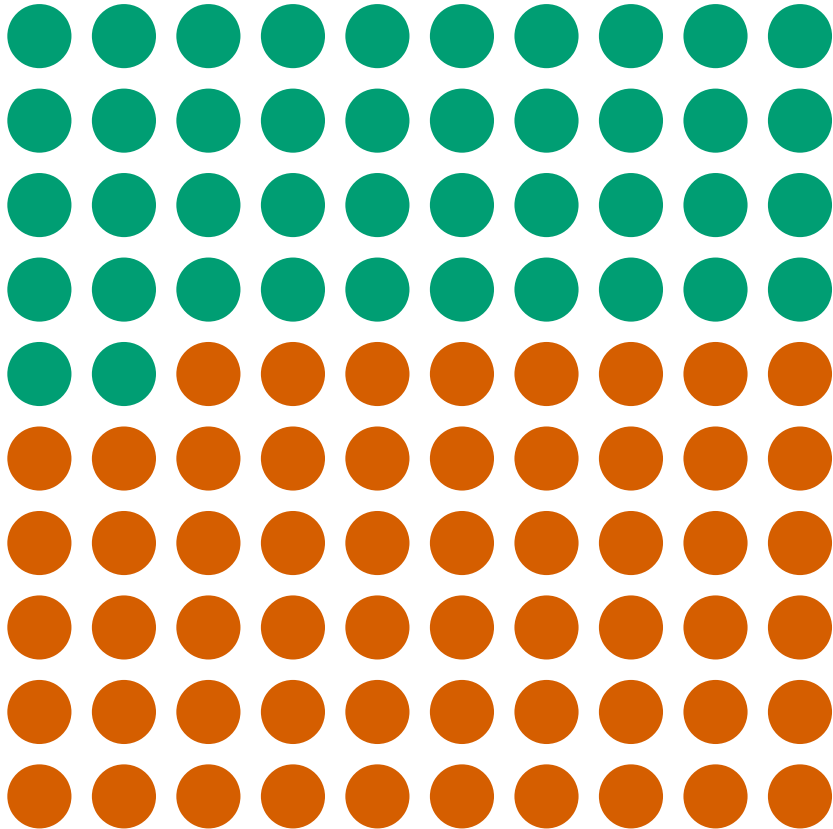
maruttuvarkaḷ paḷaiya cuvar mūlam vantārkaḷ.

the repair: only the ending corrected

Metric	Score: system output	Score: repair	Prefers
CometKiwi-22 (no reference)	0.813	0.872	the repair
COMET-22 (with reference)	0.916	0.985	the repair

On **this** pair both metrics get it right. Across **all 1,479** real errors, CometKiwi-22 ranks the correct version first only **42.3%** of the time (Result 2).

Measurement 1: direction, the "win rate"

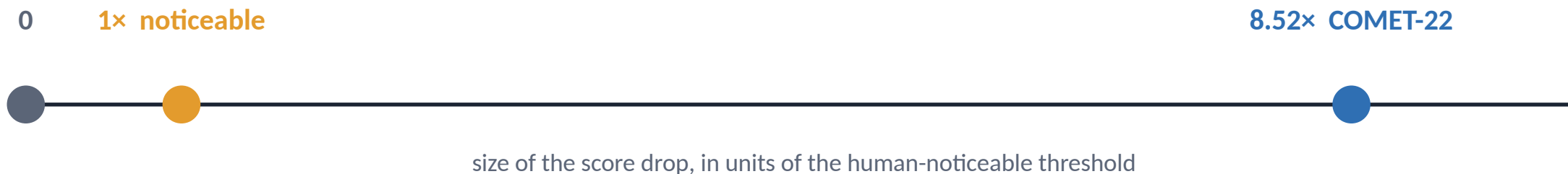


■ correct version scored higher

■ wrong version scored higher

- For every pair: does the metric score the **correct** version **higher** than the wrong one?
- **Win rate** = share of pairs where it does.
- **50% = a coin flip**. A good metric should be far above 50%.
- **Below 50%** means the metric prefers the **wrong** version more often than the right one.
- The grid shows **42 of 100**: close to what CometKiwi-22 does on real errors (42.3%).

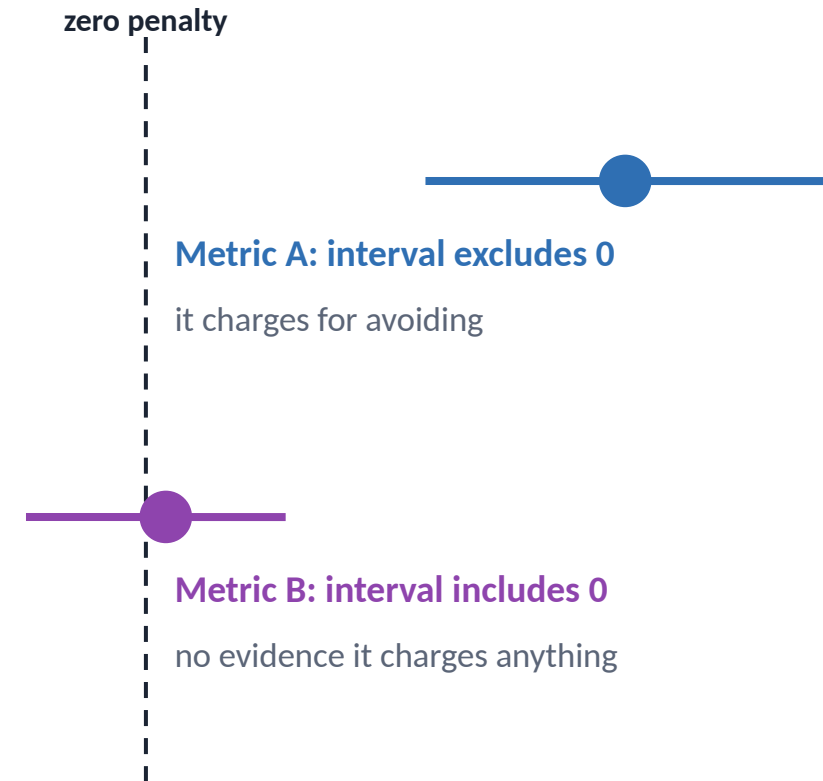
Measurement 2: size, "would a human notice this gap?"



- **The threshold** (Kocmi et al., 2024): the score difference at which human judges agree with the metric **3 times out of 4**. We call it the **75%-agreement threshold**.
- We report each penalty as a **multiple** of it: **1x** = just noticeable; **8x** = very large.
- Each metric has **its own** threshold. **3 metrics** have none published, so we show **n/a** rather than borrow another metric's.

Measurement 3: the avoidance penalty and its uncertainty

- **Avoidance penalty** = average score drop when the neutral form replaces the correctly gendered form (1,252 real outputs).
- **95% confidence interval**: the range where the true average plausibly lies. We find it by re-drawing our data at random many times and recomputing (a "bootstrap").
- **Interval includes 0** → no evidence the metric charges anything for avoiding.
- Shown as % of each metric's score range, so 0–1, 0–25 and 0–100 metrics can be compared.



(illustration)

A practical problem: our test set has no full reference sentences

The problem

Gold answers are only the **key words** (median **2 words**); the English sources are **5–9 words**. No full human reference.

Standard fix

As in **DEMETR** and **ACES**: use the correct variant of each pair as the reference.

Checking the fix

Re-score with **another system's independent translation** as the reference.



Share of pairs ranked correctly. COMET-22 stays high with an independent reference (this check was run for COMET-22's ranking).

We wrote our prediction down before seeing the results

Our prediction, frozen in advance

*"Neural metrics will **not** charge a human-noticeable penalty for a **single** obligatory morpheme."*

In plain words: "AI-based metrics will not lower the score enough for a person to notice when just one verb ending is wrong."

Pre-registration

Write the prediction down and **freeze it before** looking at the results.

Why it matters

You cannot quietly change the hypothesis afterwards ("**moving the goalposts**").

Morpheme

The smallest meaningful piece of a word. Here: the **verb ending**, e.g. -aṇa.

Statistical care, in plain words

Many tests

Running many tests makes lucky findings more likely. So we use a **stricter pass mark** (Holm–Bonferroni) inside each of **257** groups of related tests.

Similar items

Items from the same template are alike, so uncertainty is estimated by resampling **whole templates** (cluster bootstrap).

Removing an unfair bonus

Metrics prefer **unedited** text. We subtract that effect by comparing the controlled and natural arms on the same items (**de-confounding**).

The penalty sizes we defend are the **de-confounded** ones, which are **smaller** than the raw numbers.



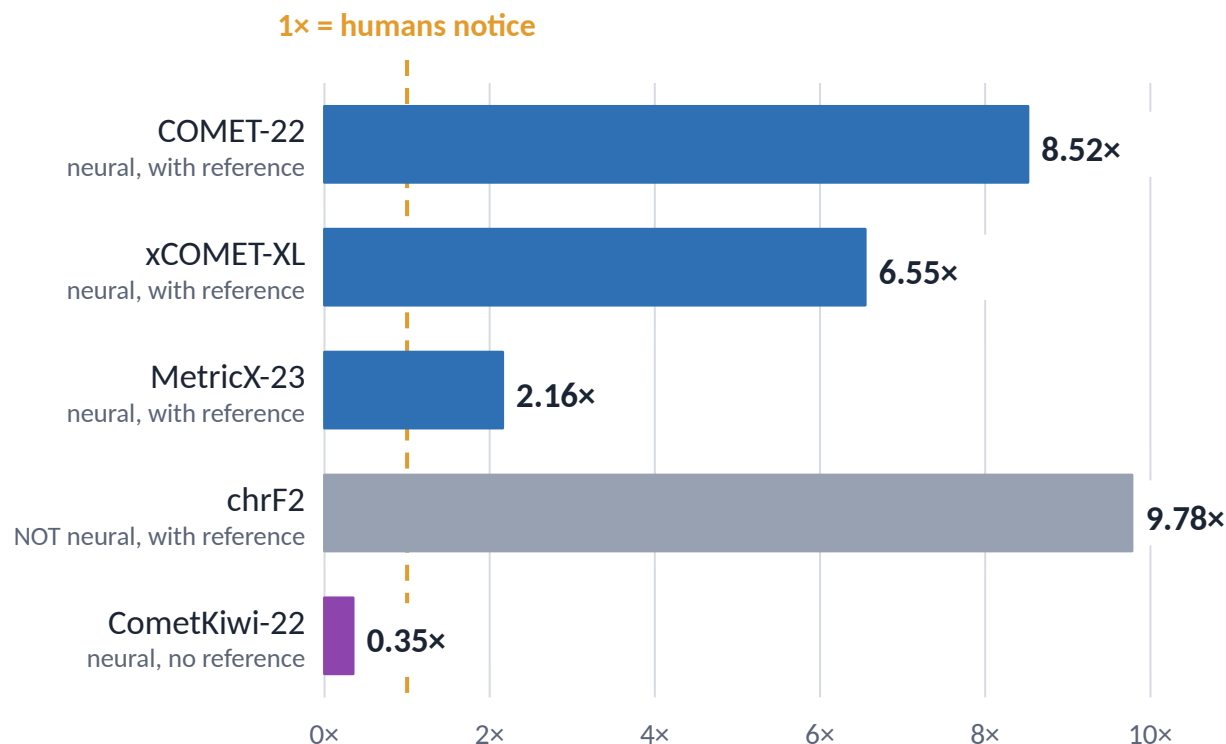
6

Results: what the metrics charge

Six findings about how metrics react to a wrong or missing value.

Result 1: our own prediction was wrong

For neural metrics with a reference: penalty for a wrong ending, in units of the human-noticeable threshold (de-confounded, from the controlled and natural arms)



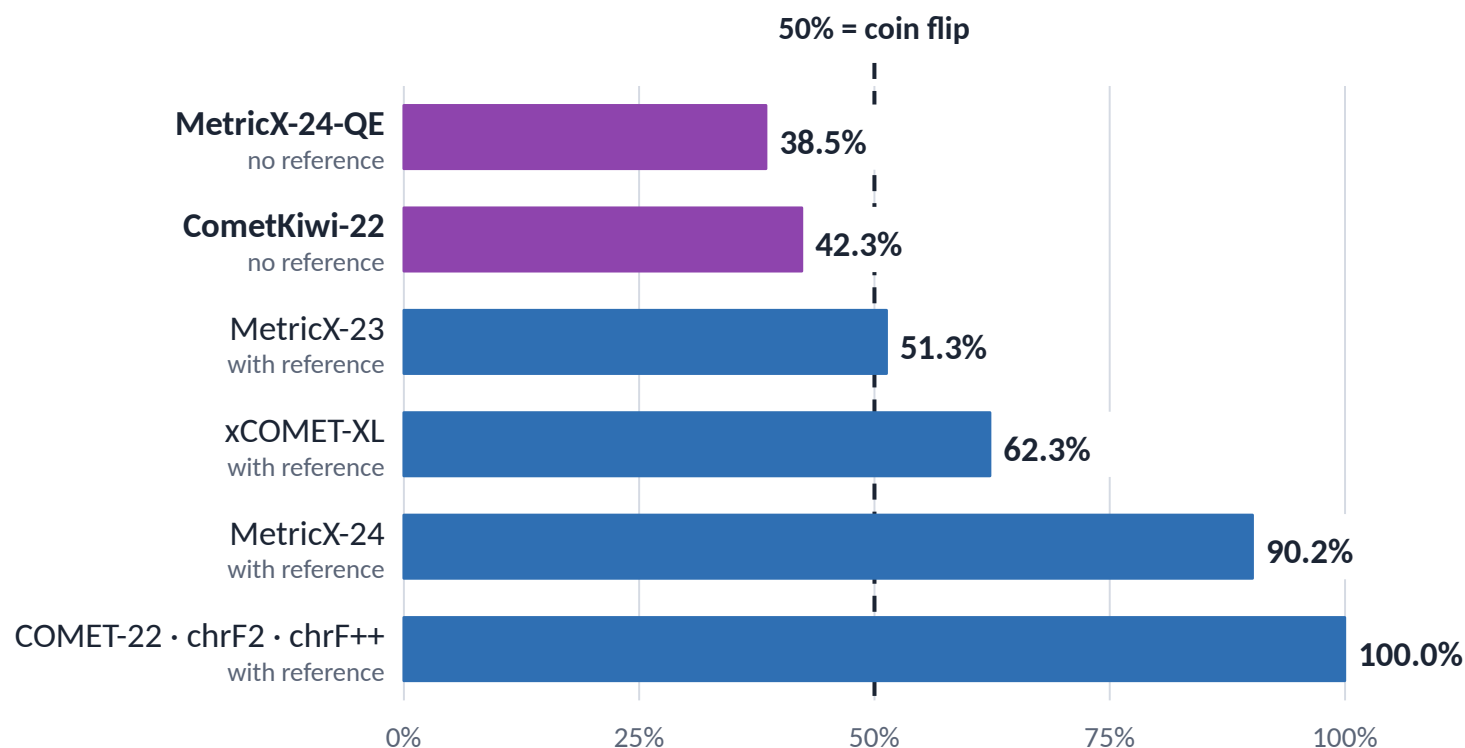
What this means

- Wrong for **3 of the 4** neural metrics that have a published threshold.
- Held only for **CometKiwi-22** (no reference).
- **chrF2** (not neural) also charges a lot, so it is not about neural metrics.
- MetricX-24, chrF++ and MetricX-24-QE: no published threshold.

"Neural metrics are blind to single-morpheme errors" is **not true** as a general statement. We report our failed prediction as failed.

Result 2: reference-free metrics prefer the wrong form

Natural arm, 1,479 real system errors: how often each metric scores the correct version higher (win rate)



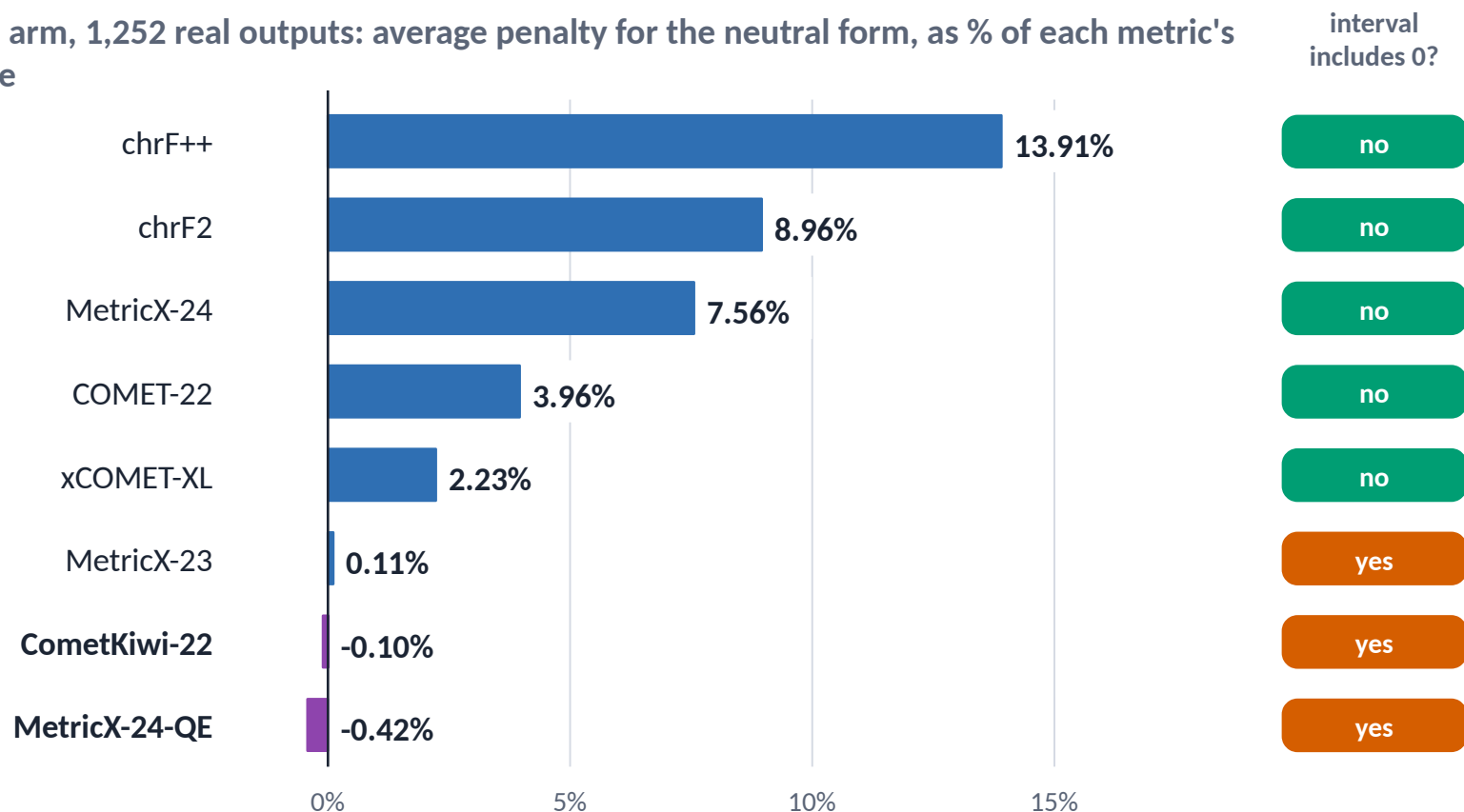
Reading the chart

- Below 50%: the metric **prefers the wrong** version.
- Both reference-free metrics are **below 50%**.
- Two metrics from **different labs** fail the same way.
- The 100% rows are partly built in: the reference is the correct version.

On the errors systems **actually** make, both reference-free metrics rank the **wrong** form first **more often than not**.

Result 3: without a reference, avoiding is free

Avoidance arm, 1,252 real outputs: average penalty for the neutral form, as % of each metric's score range



CometKiwi-22, exactly

- Average penalty **-0.000965** (on a 0 to 1 scale).
- 95% interval **-0.00426 to +0.00275**: includes zero.
- That is **0.23** of the human-noticeable threshold.
- MetricX-24-QE: interval also includes zero.

Under reference-free evaluation, a system that **never marks** the distinction **loses nothing**. Reference-based metrics do charge for it (MetricX-23 is the exception).

What "avoiding is free" looks like on one real output

English: "Akaran cried at noon." (Akaran is a masculine name) System: Gemma 3 12B

அக்காரன் நண்பகல் அழுதார்.

akkāraṇ naṇpakal aḷutār.

what the system wrote: neutral form, no gender

அக்காரன் நண்பகல் அழுதான்.

akkāraṇ naṇpakal aḷutāṇ.

the gendered version: masculine ending

Metric	Score: neutral (avoided)	Score: masculine	Penalty for avoiding?
CometKiwi-22 (no reference)	0.8844	0.8840	none (neutral even a hair higher)
COMET-22 (with reference)	0.968	0.988	yes, a clear drop

One pair in miniature; Result 3 is the same pattern over 1,252 real outputs. (Name spelled as the system wrote it.)

The paper's main table, explained column by column

Metric	Uses reference	Scale	Win on real errors (natural arm)	× noticeable (de-confounded)	Avoidance penalty (avoidance arm)	Interval includes 0?
COMET-22	yes	0–1	100.0%	8.52	3.96%	no
xCOMET-XL	yes	0–1	62.3%	6.55	2.23%	no
MetricX-24	yes	0–25	90.2%	n/a	7.56%	no
MetricX-23	yes	0–25	51.3%	2.16	0.11%	yes
chrF2	yes	0–100	100.0%	9.78	8.96%	no
chrF++	yes	0–100	100.0%	n/a	13.91%	no
CometKiwi-22	no	0–1	42.3%	0.35	–0.10%	yes
MetricX-24-QE	no	0–25	38.5%	n/a	–0.42%	yes

Bottom two rows (**no reference**): win on **under half** of real errors, and the avoidance interval **includes zero**. That is the paper's central claim.

What the paper's title means, in plain words

Obligatory Slots

→ places where Tamil forces a choice English never made

Under Reference-Free Evaluation,

→ when translations are judged without a human reference, as in most live monitoring

Dropping a Distinction the Source Never Made

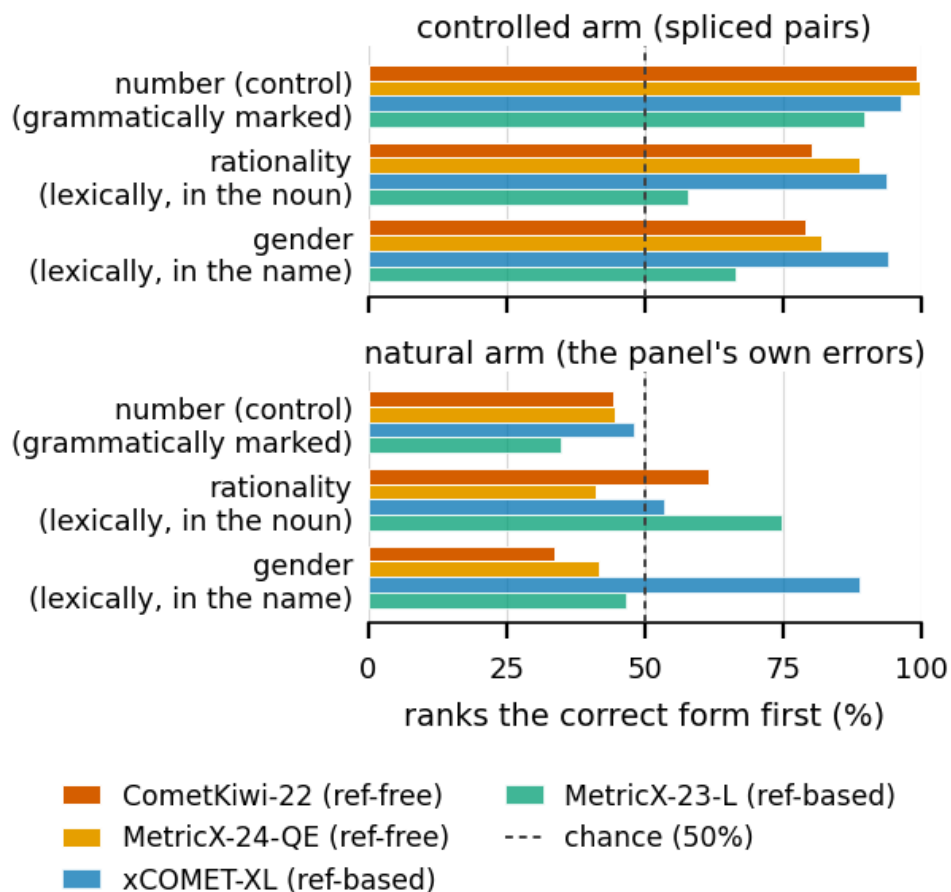
→ using a form that avoids the choice, e.g. the neutral வந்தார்

Is Free

→ the metric does not lower the score for it

The qualifier "**under reference-free evaluation**" is essential: most **reference-based** metrics **do** charge for dropping the distinction.

Result 4 (weaker): metrics rank best what English shows



Constructed pairs	number	rationality	gender
CometKiwi-22	99.2%	80.2%	79.1%
MetricX-24-QE	99.7%	88.9%	81.9%

- **Number:** English marks it in grammar (cow / cows) → ranked almost perfectly.
- **Rationality** and **gender:** English leaves them to the **meaning** of the noun or the name → ranked worse.
- So a metric ranks best what the English source makes **easiest to recover**.

Paper Figure 1. Top: constructed ("spliced") pairs. Bottom: real errors ("the panel" = our 13 systems). Dashed line = 50% coin flip.

Result 4: three honest qualifications

1 • Only in constructed pairs

- On **real errors** the pattern disappears.
- CometKiwi: **44.3%** on number vs **61.6%** on rationality.

2 • Not only reference-free

- MetricX-23 (with reference) separates the slots **more**: 89.8% vs 57.9%.
- xCOMET-XL (with reference) is almost flat: 96.5% vs 93.9%.

3 • Strongest part withdrawn

- The lowest scores were on two slots English cannot mark at all (CometKiwi **67.9%** and **20.7%**).
- Both slots were **withdrawn** for construction flaws.

Result 4 is **suggestive, not central**. The central claim is Results 2 and 3, which hold on **real outputs**.

Result 5: xCOMET finds the area of the error, not the error

One sentence (illustration of average proportions)

the actual wrong ending: ~3.3 characters



area xCOMET flags: 71.2% of the sentence

91.1%

of items: the flagged area contains the ending

9.4%

of items: flagged area actually matches the ending ($\geq$ half overlap)

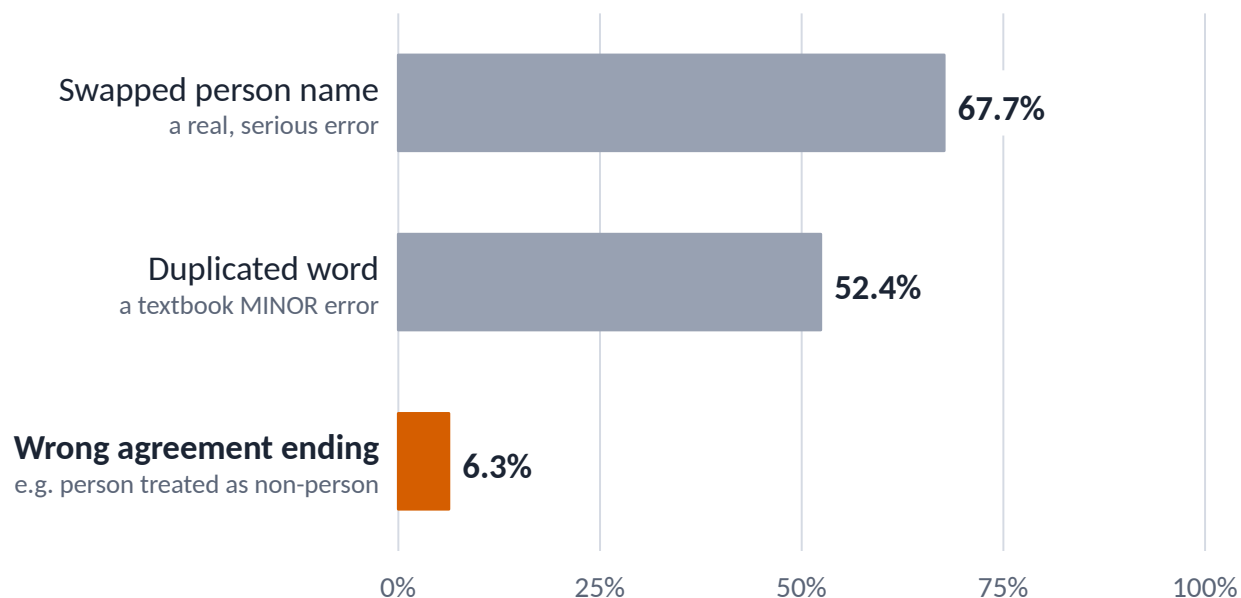
72.9%

of correct, unchanged sentences also get flagged

xCOMET **detects** that something is wrong somewhere, but **does not localise** the error.

Result 6: xCOMET rates the error as minor

When xCOMET flags the change, how often does it call it "critical"? (same sentences)



Why this is convincing

The comparison errors are placed in the **same sentences**, so "xCOMET is just weak on Tamil" cannot explain it.

A duplicated word is rated critical **about 8× more often** than a wrong agreement ending, which can put a person into the non-people class.

Why not just retrain the metric?

What earlier retraining did

- Showed **how far** retraining can close a blind spot, for features **marked in the source**.
- Training examples could be **generated automatically**: change the marked feature, and the right answer is known.

Why it is harder here

- The source **does not say** the value.
- So the automatic recipe for making training examples **does not apply**.

Earlier retraining worked only when the English **showed** the feature (Amrhein & Sennrich 2022; Singh et al. 2024). Here it does not.



7

Results: what the systems do

Four findings about how the 13 systems behave. These are provisional; the caveats are stated with them.

Result 7: can systems produce the form when told to?

System	Slot	Original instruction	Fixed instruction
Gemma 3 27B	gender	80.0%	100.0%
Gemma 3 27B	number	50.0%	100.0%
Qwen3 8B	number	35.0%	100.0%
Qwen3 32B	gender	65.0%	100.0%
Gemma 3 12B	rationality	50.0%	50.0%

A few rows of the paper's Table 8 (accuracy in the instructed condition).

What we learned

- The flaw: it said "realising the rationality value 'UYAR' on the verb": an **internal label code**, not the Tamil form.
- Fixing it moves accuracy by **up to 75 points**.
- Even fixed: **6 of 11** beat chance on gender, **0 of 11** on rationality (the 2 MT systems take no instructions).

If a system scores low on rationality, we **cannot tell** whether it chose wrongly or simply **cannot write that form** on request. Both instruction versions are reported.

Result 8: context makes systems commit, not succeed

Rationality · commitment (share of items where a value was chosen), range across the 13 systems

↑ rises on 12 of 13

In-sentence clue



Prior-sentence clue



Rationality · accuracy among committed items, range across the 13 systems

↓ falls

In-sentence clue

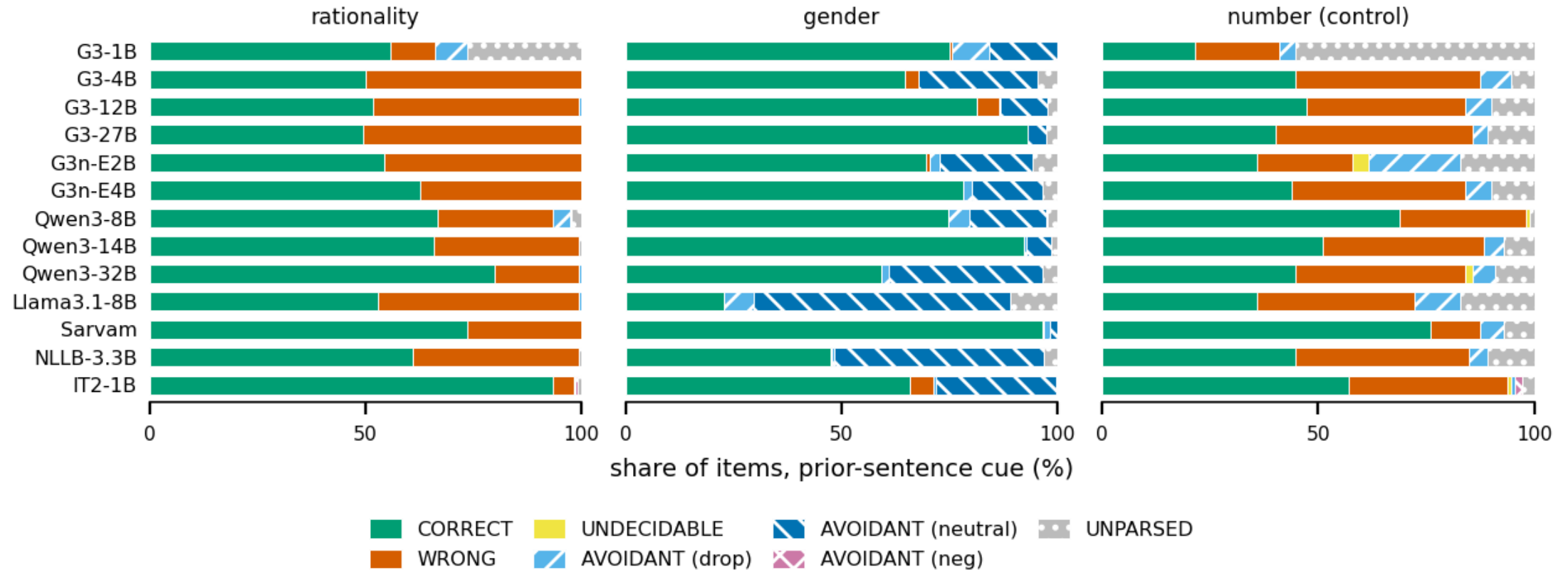


Prior-sentence clue



The context sentence **stops systems avoiding** and makes them **commit anyway**, often wrongly. Holds across 4 model families and both MT systems. Caveat: one prompt only.

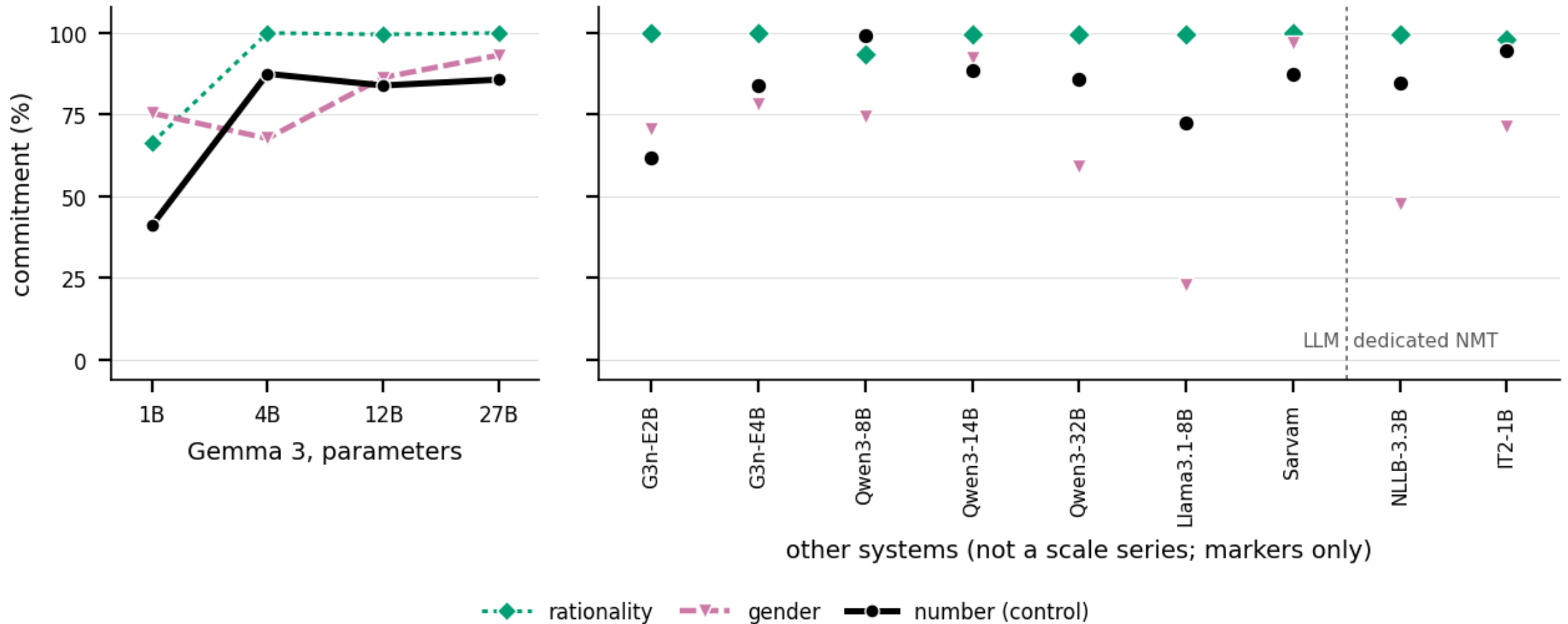
What each system does with a prior-sentence clue



G3 = Gemma 3 · G3n = Gemma 3n · IT2 = IndicTrans2 · Sarvam = Sarvam-Translate. Green = correct, orange = wrong, blue hatched = avoidant, grey dotted = unparsed.

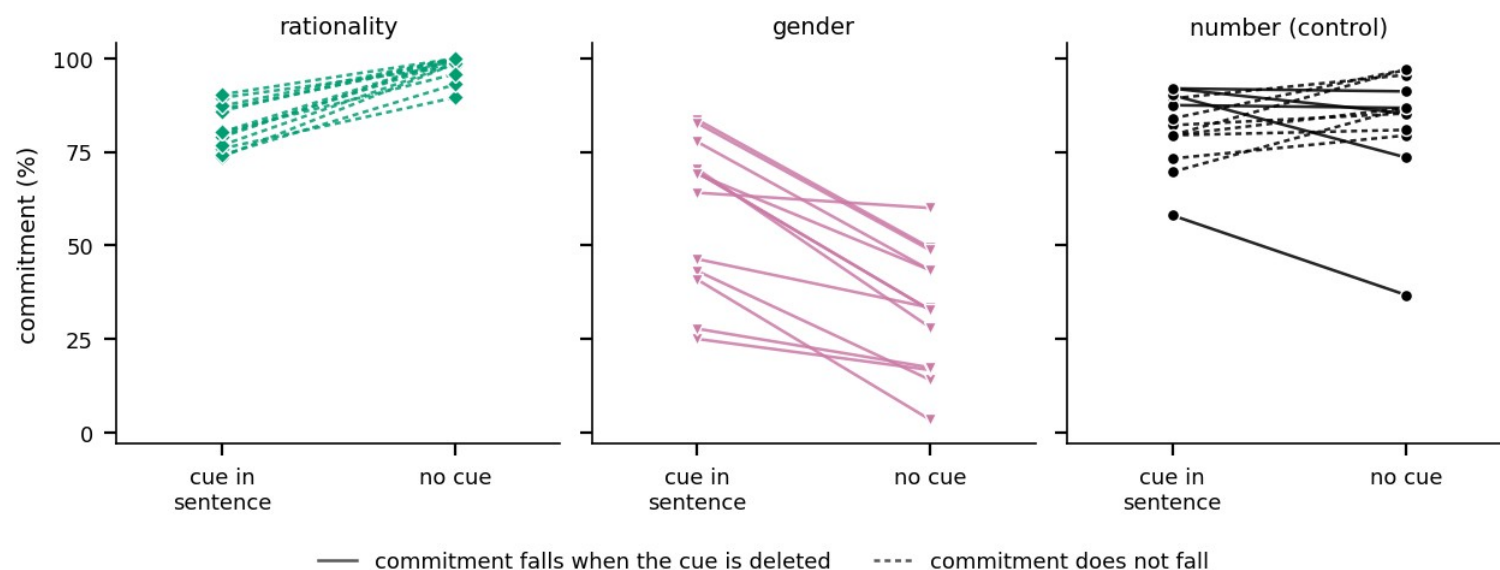
Rationality: nearly everyone **commits**, often **wrongly**. Gender: much **avoiding** (blue), e.g. Llama 3.1 8B avoids most of the time.

Commitment per system with a prior-sentence clue



G3n = Gemma 3n · IT2 = IndicTrans2 · Sarvam = Sarvam-Translate · "dedicated NMT" = the two dedicated machine-translation systems.

Result 9: when the clue is deleted



Each line is one system: commitment with the clue in the sentence (left end) and with no clue (right end). Solid = falls, dashed = does not fall.

Gender

13 of 13

falls on every system (by 4.1–42.4 points)

Rationality

0 of 13

never falls: systems answer as confidently without a clue

Number

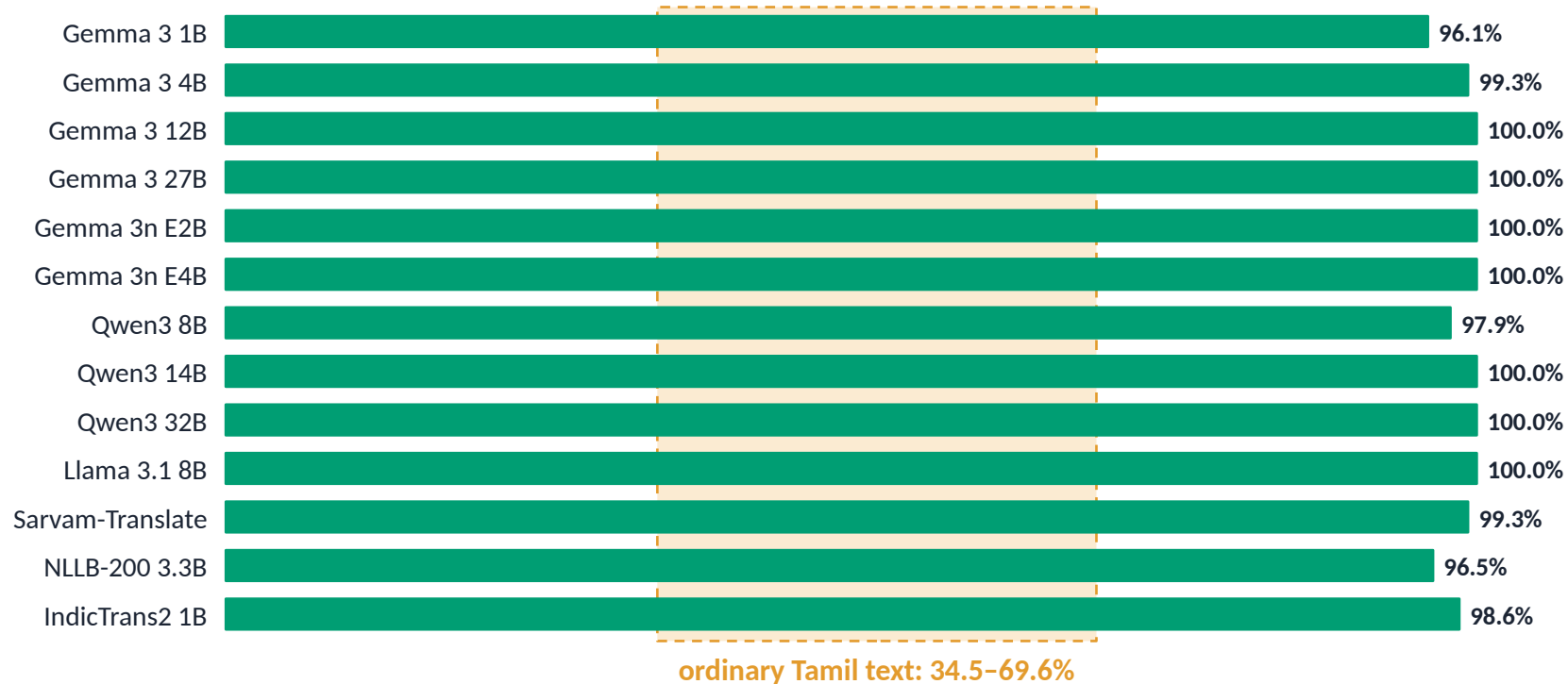
5 of 13

not interpreted (checker issue in this slot)

Without a clue, systems **avoid** in gender, but **keep choosing** in rationality. So what do they choose? Next slide.

Result 10: with no clue, every system picks "people"

Cue-free rationality: share of committed outputs using the people form (உயர்திணை)



Gender

Among systems with enough data:
5 default to masculine, 3 to feminine.

The systems share a strong default toward **"people"**, far stronger than in ordinary Tamil text. Why is unknown: no system's training data was measured.

What we do NOT claim about the systems

No comparison between slots

- The number control does not work on these items: accuracy **46.9–86.7%** vs chance 50%.
- The analyser reads some number gold unevenly.

Gender: no direction

Between the two clue positions, gender shows **no reliable direction**.

One prompt only

The prior-sentence condition used **one** prompt, so clue position and prompt wording **cannot be separated**.

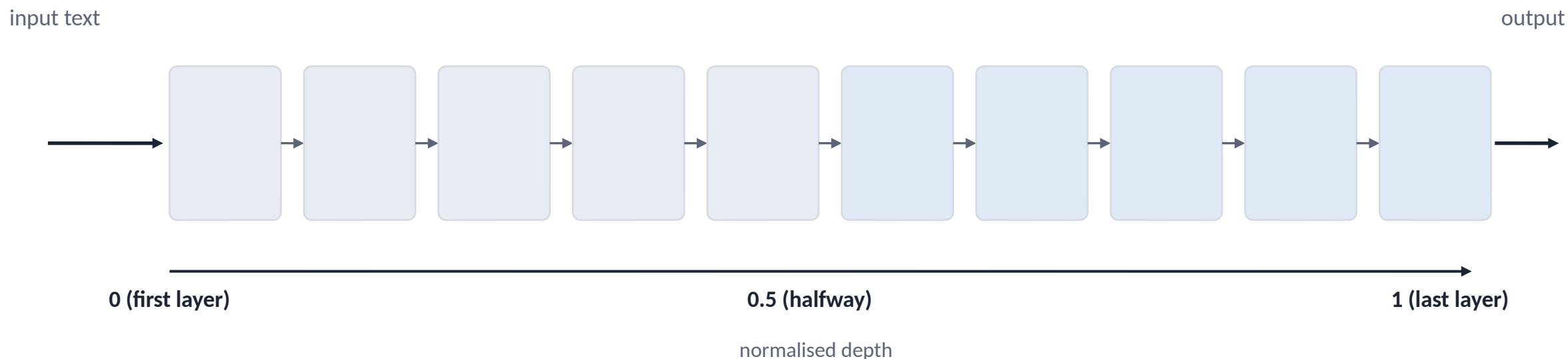
That is why the paper calls this section "**Model Behaviour, Provisionally**".

8

Inside one model: where is the decision made?

A pre-registered experiment on gemma-3-4b-pt, reported together with the part that failed.

First, what "depth" inside a model means



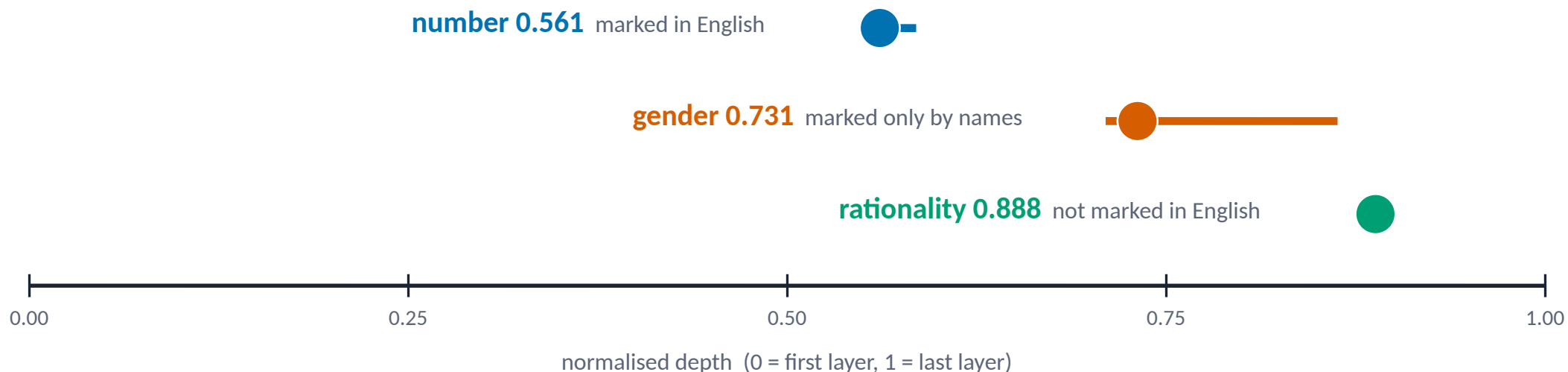
Why ask? A reference-free metric cannot see a distinction the English does not mark. Does the **model itself** also settle it late? Model studied: **gemma-3-4b-pt** (Gemma 3 4B, before chat training).

Activation patching, in simple words



- If the "dogs" run now produces the **people** ending, that layer **already carried the decision**.
- Repeat at **every depth**, in **both directions** (students→dogs and dogs→students).
- **Onset** = the first layer where pasting moves the answer **at least halfway** towards the other answer, and keeps doing so at every later layer.

Result 11: rationality is settled much later than number



Slot	Marked in English?	Onset depth (95% interval)
number (control)	yes	0.561 (0.550–0.585)
gender	no (only via names)	0.731 (0.710–0.863)
rationality	no	0.888 (0.878–0.891)
gap: rationality – number		0.327 (0.301–0.339)

Checks passed

- The gap is clearly above zero and above the **0.200** minimum set in advance.
- Both directions agree; the number control works (**0.992**).

What failed: the probe study, reported anyway

What a probe is

A small classifier trained to **read a feature** (e.g. rationality) straight from a layer's internal state.

Why the main result still stands

The frozen plan made **patching the primary** method and the probe secondary. Patching shows **where** the decision is causally fixed.

How much better the probe did than a simple word-based baseline (AUC gain; AUC 0.5 = guessing, 1.0 = perfect)



What the failure costs

No claim about **when** the feature becomes readable. A further test that could tell "not there yet" from "there but hidden" was **not run**.

The probe missed the bar we set in advance, so we **dropped it instead of lowering the bar**.

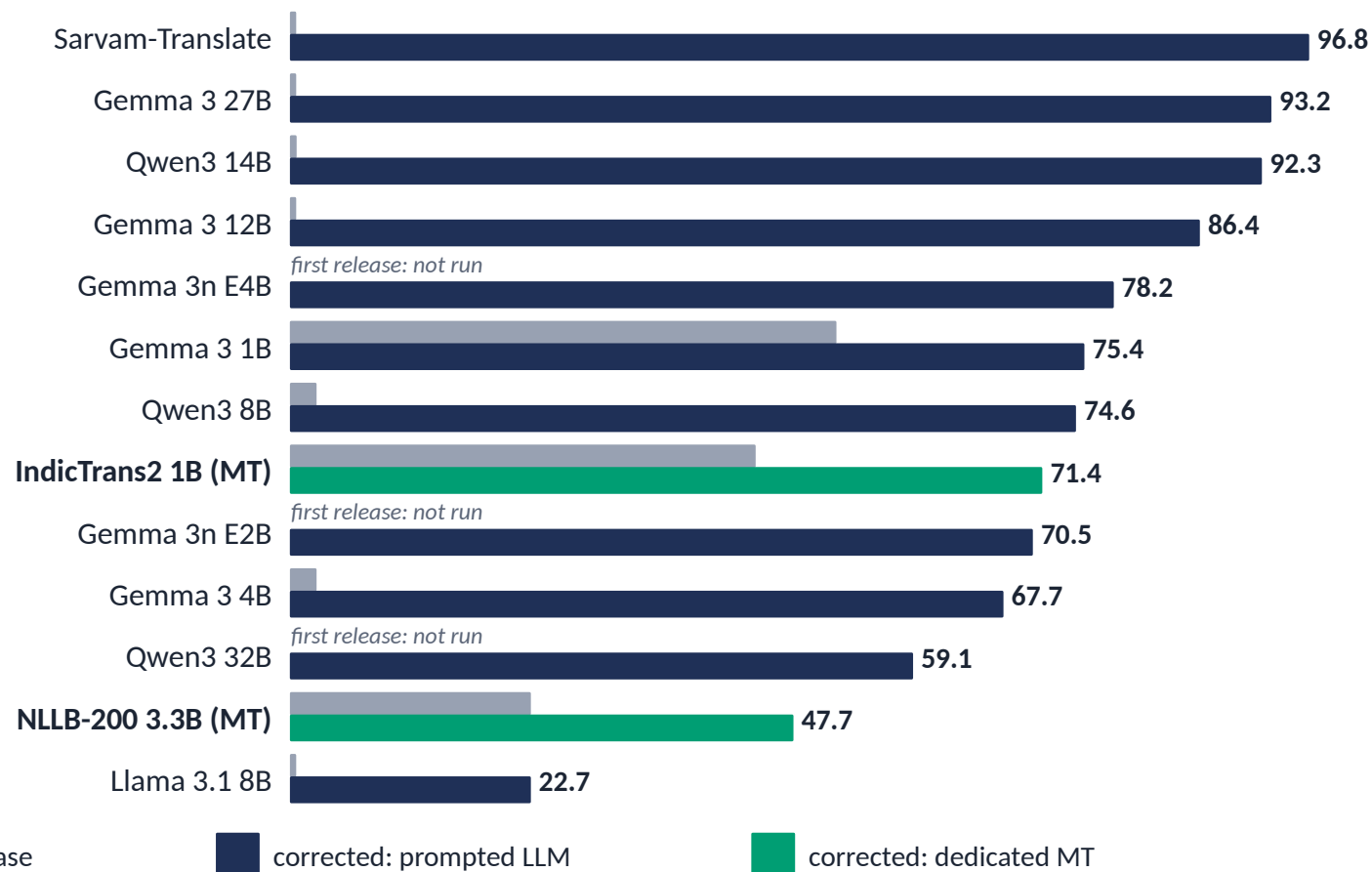
9

A mistake we caught in our own first release

A clean, quotable result that turned out to be built into the test items. We retracted it.

The claim we retracted: it did not just weaken, it reversed

Prior-sentence gender commitment (%)



What happened

- First release: **most** language models almost never committed; the MT systems did.
- Corrected: language-model median **75.4%** vs MT **47.7 to 71.4%**; **7 of 11** exceed both MT systems.
- The result did not just shrink: it **flipped**.

Three independent flaws caused it

1 • A neutral pronoun

- Items used singular "**they**" for one woman.
- Tamil has a neutral form, so a neutral translation was **correct**, but was scored as avoidance.

2 • Different kinds of clue

- In-sentence: guess gender from a Tamil **name** (world knowledge).
- Prior-sentence: stated **in plain English** (reading).
- So the "harder" condition was easier on **11 of 13** systems.

3 • A weak check for translating the wrong sentence

- It checked only **output length**.
- Missed systems translating the **context** sentence instead of the target.
- Gemma 3 1B missed the target verb on **98.2%** of these items (38.6% with an in-sentence clue).

The three flaws are **independent**. Because of flaw 2, **no gender comparison** between the two clue positions is drawn anywhere.

The simple audit that caught them, now run on every build

The audit asks one question per slot and condition: **does the English source contain any word that expresses the feature the gold answer commits to?**

1,141

unanswerable items in the first release, across 12 categories

0

in the corrected release, where it now stops the build

- **Deliberately simple:** word lists, no clever parsing, so every decision can be checked by hand.
- We **keep** the flawed first version as a test, so every rebuild checks that the audit **still catches** its known errors.

A second check: re-scoring after the analyser repair

54,443

records re-scored

728

outcomes changed (725 in
rationality)

717

became correct (512 unparsed,
139 wrong)

0

metric-study errors that turned
out not to be errors

In-sentence rationality commitment	Before repair	After repair
Range across 13 systems	58.3–75.0%	73.7–90.3%

The lesson

One analyser flaw moved a headline number by up to **21.1 points**. We publish the scores **before and after**.

The old analyser was **under-counting commitment**, not inventing errors. All metric results are **unchanged**.

Native Tamil speakers corrected the data

Ruling 1 • "You" has two levels (the removed slot)

In statements: நீ / நீர் (less formal) vs நீங்கள் / தாங்கள் (polite). **Two levels.**

Ruling 2 • Plural non-people needs the plural ending

நாய்கள் வந்தன

nāykaḷ vantaṇa

✓ correct

நாய்கள் வந்தது

nāykaḷ vantatu

✗ wrong (singular)

The general lesson

A rule-based tool can say a word **follows its rules**, not that Tamil speakers **really use it**. Two forms that passed its round-trip check had **no occurrences** in our Tamil text collections. Verification is **human-led, with the tool as support**.

10

Limits and conclusion

What we can say, what we cannot, and what to take away.

Limitations, stated plainly

One language

written Tamil, one agreement system; may not generalise

No frontier systems

open models only; the largest has 32B parameters

Number control withheld

the analyser reads only 498 of 580 number gold forms, unevenly

One prior-sentence prompt

clue position and prompt wording cannot be separated

Analyser sensitivity

one repair moved a number by up to 21.1 points

Probe failed

no claim about when a feature becomes readable

Also worth knowing

the "human-noticeable" threshold is borrowed from Kocmi et al. (2024); no human study was run on these items

What this means in practice

Reference-free monitoring alone

Will **largely miss** these errors, and will **not penalise** a system that skips the choice.

Reference-based metrics

Do charge for these errors, but need **human references**, which live systems rarely have.

A targeted check helps

A rule-based check of the verb ending, like the one in this benchmark, is built to read exactly these endings (though it cannot read every output).

Do not trust severity labels here

xCOMET rarely calls these errors **critical** (6.3%), even when it flags them.

These are **our reading** of the results, not separate experiments.

Conclusion

Our prediction was wrong

for neural reference-based metrics: they **do** charge, e.g. COMET-22 at 8.52× the noticeable threshold.

Reference-free metrics prefer the wrong form

on real system errors: CometKiwi-22 **42.3%**, MetricX-24-QE **38.5%** (coin flip = 50%).

Skipping the choice is free

under reference-free evaluation: both avoidance intervals **include zero**.

The systems

with no clue, every system picks the **people** form on 96–100% of rationality items.

Inside one model

number is settled at depth **0.561**, rationality only at **0.888**.

Under reference-free evaluation, dropping a distinction the source never made is free.

Key numbers at a glance

8.52×

COMET-22 penalty, in units of the human-noticeable threshold

42.3%

CometKiwi-22 win rate on 1,479 real errors (coin flip = 50%)

38.5%

MetricX-24-QE win rate on the same errors

−0.000965

CometKiwi-22 average penalty for skipping the choice, on a 0–1 scale (interval includes 0)

6.3%

how often xCOMET calls the agreement error "critical" (duplicated word: 52.4%)

0.561 → 0.888

depth where number vs rationality is settled inside gemma-3-4b-pt

1,141 → 0

unanswerable items: first release vs corrected release

13

translation systems tested; 2,873 items in the release

Glossary (1 of 2): every technical word, in one line

Agreement	the verb changes its ending to match its subject	Morpheme	the smallest meaningful part of a word, e.g. a verb ending
Rationality (tiṇai)	Tamil class: people and gods vs everything else	Marked / unmarked	shown / not shown by a word or ending in the sentence
Obligatory slot	a value the Tamil must show but the English does not give	Value	what fills the slot, e.g. "people" or "not people"
Neutral form (வந்தார்)	correct Tamil that does not say male or female	Cue	a clue in the text that reveals the right value
Item	one test sentence plus its correct answer	Gold	the correct answer for an item
Template	a sentence pattern that items are generated from	Control	a part of the test where the expected result is already known
Commitment	share of items where the system chose a value	Accuracy	among committed items, the share that were right
AVOIDANT	the output chose no value (neutral form, dropped verb, negation)	UNPARSED	the analyser could not read the output
UNDECIDABLE	the form stays ambiguous even with context	FST analyser	rule-based program that reads Tamil word forms

Glossary (2 of 2): every technical word, in one line

Language model (LLM)	large neural network that continues text; translates when asked	Prompt	the instruction given to a language model
Parameters (4B ...)	learned numbers inside a model; 4B ≈ 4 billion	Fine-tuning	extra training on one narrow task
Dedicated MT system	a network built only to translate; needs no prompt	Neural	built from a trained neural network, not hand-written rules
Reference-based metric	scores a translation against a human reference	Reference-free (QE)	scores using only the source and the output
Arm	one kind of evidence: controlled, natural or avoidance	Win rate	share of pairs where the correct version scores higher
75%-agreement threshold	score gap at which humans agree with the metric 3 times in 4	Confidence interval	the range where the true value plausibly lies
Bootstrap	re-drawing the data at random many times to measure uncertainty	De-confounding	removing the metrics' preference for unedited text
Pre-registration	freezing the prediction before seeing the results	Corpus prior	how often a value appears in ordinary Tamil text
Activation patching	pasting a model's internal state from one input into another	Onset (depth)	first layer where the decision is settled; 0 = first, 1 = last

Thank you

Questions are welcome.

*Under reference-free evaluation,
dropping a distinction the source never made is free.*

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TamilLingBench data (CC BY-SA 4.0) and code (Apache-2.0) are released with the paper.

Backup: a practical lesson for others doing this kind of work

float16

- Largest value it can hold: **65,504**.
- Gemma 3's internal values **exceed** it in the upper third of the network: exactly where this result lives.

bfloat16

- Matched **197 of 200** predictions.
- Error up to **0.25**: larger than several effects we measure.

float32 (used)

- Matched **200 of 200** predictions.
- Largest error **0.0000467**.

If you patch activations in Gemma 3, **use float32**. A 16-bit format can silently corrupt exactly the layers you care about.